Lost in translation or found in analogy?

Semantics d: K -> Q and p: K -> V create a bridge:
Q <- K -> V

Sections c: Q -> K and s: V -> K of d and p respectively, create a handshake:
Q -> K <- V

Interpretation:
Let Q denote a foreign language, V denote its translation into a home language, then K is an internal machine representation of both. p is like a dictionary (s look up) and d a foreign language dictionary (c look up). Notice the home inductive bias. (Data: queries qi ε Q, i ε [n] and key-value pairs (kj, vj) ε K x V, j ε [m]) The semantic handshake is literally where meaning can get lost in translation. (In K, query meaning c(qi) is only in proximity to value meaning s(vj*))

The degree to which the diagonal fibration liftings between two languages approximately commute, reflects their degree of linguistic similarity. Moreover, the fidelity of amortization is a quantitative measure of meaning lost translation.

Potential Chu-former applications of Q&A and Translation:
1. Interpretation
• Involves conveying spoken or signed communication between users of different languages.
• Focuses on real-time understanding and response.
2. Localization
• Adapting content to meet the cultural and linguistic needs of a specific audience.
• Goes beyond translation to include cultural nuances and local customs.
3. Transcription
• Converting spoken language into written text.
• Often used in legal, medical, and academic settings.
4. Subtitling
• Adding translated text to video content.
• Requires synchronization with audio and visual elements.
5. Paraphrasing
• Restating text or speech in different words while maintaining the original meaning.
• Useful for simplifying complex information.
6. Summarization
• Condensing information into a shorter form.
• Focuses on key points and main ideas.
7. Adaptation
• Modifying content to fit a different context or medium.
• Common in literature and film, where stories are reimagined for new audiences.

These uses share similarities with Q&A and translation in their focus on communication and understanding across different languages and formats.

8. Differential Diagnosis
More generally, in terms of abductive reasoning, view K as a set of hypotheses and V as a set of manifestations related by an effect function p: K -> V. Using a fibration lifting, a function of d and p, find a section of effect p, that is s: V -> K satisfying p(s(v)) = v, which says s(v) explains v under semantics p; a form of abductive reasoning. Similarly for effect d: K -> Q. d and p create a semantic bridge Q <- K -> V and their sections a handshake.

Note that the sections of d and p depend on each other through the fibration liftings, making the handshake coherent.

Now let pairs (kj, vj), j ε [m], denote (disease, symptom), and let a patient's symptoms Q = {v1, … vl } be a subset of symptoms {v1, … , vm} where l < m. Let d: K -> Q, d(kj) = vj, j ε [l], be the current physician diagnosis and p: K -> V, p(kj) = vj, j ε [m], be possible differential diagnoses from medical literature. Sections c and s of d and p respectively, found from fibration liftings involving d and p, form an abductive handshake. Indeed for j ε [l], each patient disease c(vj) is in proximity to a cloud of possible diseases s(vj'), j' ε [m]. Choose a few of the closest diseases. This generates a superset of potential patient diseases with symptoms suggested by differential diagnosis p. Further testing can rule out or confirm a differential diagnosis.